

NMAM INSTITUTE OF TECHNOLOGY, NITTE*(An Autonomous Institution affiliated to VTU, Belagavi)***Seventh Semester B.E. (ISE) (Credit System) Degree Examinations**

January 2023

19ISE52 – NETWORK ENGINEERING

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Explain the purpose of using TCP window scaling also show the option field format when TCP window scaling is enabled.	10	L*3	2	3
b)	Consider the initial congestion window is 10 segments (where 1segment=1460bytes), RTT=70ms, calculate the time taken to transmit 64KB.	6	L2	1	2
c)	Differentiate between Internet Draft and RFC	4	L2	1	1
2. a)	Explain the working of Fast Retransmit algorithm with an activity diagram, List the limitations and mention the potential solutions proposed	10	L2	2	3
b)	Explain TCP's three-way handshake mechanism for connection establishment and provide the optimization proposed to avoid TCP handshake for minimize the latency.	10	L2	2	3
3. a)	Explain TCP's slow start and AIMD algorithm with a neat diagram. Mention the impact of AIMD algorithm and specify the proposed TCP variants to solve the issues related to AIMD.	10	L3	1	2
b)	Explain how TCP Newreno handles multiple packets drops from the same window and yields better throughput compared to TCP Reno.	10	L3	1	2
Unit – II					
4. a)	Explain rate halving technique with SACK and without SACK using activity diagram	10	L2	2	2
b)	Develop an example timeline for SSRB when pipe > ssthresh and pipe ≤ ssthresh using the given data as follows: pipe = 4 segments, RecoverFS = 10 segments and ssthresh = 5 segments	10	L2	2	2
5. a)	Mention the components of RED and Explain the working of RED algorithm with the pseudo code	10	L3	3	2
b)	Explain router architecture with a neat diagram showing the components of a router	6	L3	3	2
c)	Differentiate between adaptive RED and self-configuring RED	4	L3	3	2
6. a)	Suppose the target range defined in adaptive RED is modified with the value 0.48 and 0.52, calculate the new bounds for α and β	10	L3	3	3
b)	Describe Explicit Congestion Notification (ECN) codepoints in TCP header and IP header. Also draw ECN negotiation diagram	10	L2	4	2
Unit – III					
7. a)	Explain the working of CoDel in dropping and non-dropping phases. Also mention the known problem with CoDel	10	L3	3	3
b)	Explain the working of Proportional Integral (PI) controller and PI controller Enhanced (PIE) Queue disciplines. Show the calculation of drop probability in each qdiscs.	10	L3	3	3
8. a)	Describe the working of Data center TCP (DCTCP) by showing the modification required at the sender, receiver and switch side.	10	L3	5	2
b)	Brief on TCP incast and TCP outcast challenges for TCP in Data Center Networks with relevant diagrams.	10	L3	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
